Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.Tech ECE - V
Name of the Paper: Analog and Digital Communication
Paper Code: TEC 501

Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.

Q1. (10 Marks)

a. Explain the demodulation of SSB-SC? Derive the mathematical expression for it? **CO1**

OR

b. What is Hilbert Transform? Explain all the properties of Hilbert Transform mathematically? **CO1**

Q2. (10 Marks)

a. Describe the process of detection of DSB-SC? Explain it mathematically? **CO1**

OR

b. Derive the formula for efficiency of AM wave showing its relationship with modulation index? If the value of modulation index is given 0.5,0.707 and 1 find out the values of the efficiencies for these values? **CO1**

Q3. (10 Marks)

a. Explain the generation of DSB-SC method using Ring Modulator with proper diagram? **CO1**

OR

b. If modulation index of a single tone AM signal is given 0.7 and 1. Find out the values of E_{max} and E_{min} for both the values? Carrier amplitude is given 5 V? **CO1**

Q4. (10 Marks)

a. What is instantaneous frequency? Determine and draw the instantaneous frequency of a wave having a total phase angle given by, $\Psi(t) = 2000t + 10\,t$. **CO2**

OR

b. Explain the generation and detection of Phase modulation? **CO2**

Q5. (10 Marks)

a. Explain the generation and detection of Frequency Modulation? **CO2**

OR

b. A single tone modulating signal $\cos(15\,\pi\,10^3\,t)$ frequency modulates a carrier of 10 MHz and produces a frequency deviation of 75 KHz. Find (i) the modulation index and (ii) phase deviation produced in the FM wave. (iii) If another modulating signal produces a modulation index of 100 while maintaining the same deviation, find the frequency and amplitude of the modulating signal, assuming $k_f = 15$ kHz per volt. **CO2**